



Essential Git Commands

A practical reference for everyday Git workflows — from resetting your working tree and syncing with remotes, to configuring your identity and managing branches. Designed for developers who live in the terminal.

[GIT](#)[CLI REFERENCE](#)[DEVELOPER TOOLS](#)

Reset, Clean & Sync Your Repository

When your working directory is messy or out of sync with the remote, these commands help you start fresh and pull in the latest changes safely. Use with care — `git reset --hard` discards uncommitted changes permanently.

`git clean -fd`

Removes all untracked files and directories from the working tree. The `-f` flag forces deletion; `-d` includes directories.

`git reset --hard HEAD`

Resets the index and working tree to match HEAD. All staged and unstaged changes are discarded. Use only when you are certain you want to lose local modifications.

`git pull origin`

Fetches changes from the origin remote and merges them into your current branch. Run after a hard reset to bring your local branch fully up to date.



Warning: `git reset --hard HEAD` permanently deletes uncommitted changes. Always verify you do not need those changes before running this command.

Fetching & Pulling from All Remotes

Use `git fetch` and `git pull` with the `--all` flag when your repository has multiple remotes configured. These commands ensure you are synchronising with every remote, not just origin.

`git fetch --all`

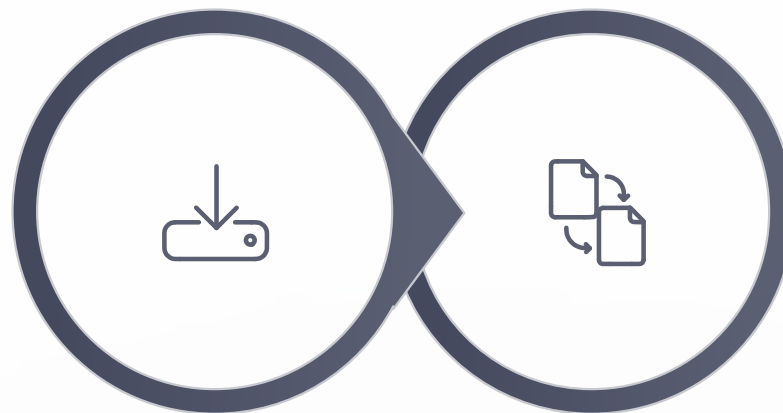
Downloads all objects and refs from every configured remote without merging. Your local branches remain untouched — this is a safe, non-destructive operation.

- Updates remote-tracking branches
- Does not modify your working tree
- Safe to run at any time

`git pull --all`

Fetches from all remotes and attempts to merge changes into your current branch. Use with caution — merging from multiple remotes can introduce conflicts.

- Combines fetch + merge across remotes
- May trigger merge conflicts
- Best used when you control all remotes



`git fetch --all`

`git pull --all`

Always run `git fetch --all` first to inspect incoming changes before merging with `git pull --all`.

Configure Your Git Identity & Inspect Remotes

Git requires a name and email to tag your commits. Set these globally once, and they will apply to all repositories on your machine. Use the inspection commands below to verify your setup and audit your remote connections.

Set Your Name

```
git config --global user.name "Your Name"
```

Assigns your display name to all future commits across every repository.

Set Your Email

```
git config --global user.email "you@example.com"
```

Links your commits to an email address. This is what GitHub and GitLab use to attribute contributions.

Verify Configuration

```
git config --global user.name  
git config --global user.email
```

Prints the currently configured values without the --global flag to check per-repository overrides.

Credential Manager

```
git credential-manager version
```

Checks the installed version of Git Credential Manager, which handles authentication for HTTPS remotes.

Remote Management & Branch Switching

Manage your remote connections and navigate between branches efficiently. These commands let you inspect, add, or remove remotes, and switch your working context to any branch in the repository.



`git remote show origin`

Displays detailed information about the `origin` remote — including tracked branches, fetch/push URLs, and whether your local branches are ahead or behind.



`git fetch --all`

Refreshes all remote-tracking references before inspecting or switching branches. Ensures you see the latest state of every remote branch.



`git remote add / remove origin`

`git remote remove origin` deletes the existing remote. `git remote add origin <url>` sets a new one. Useful when migrating repositories or fixing misconfigured remotes.



`git switch <branch-name>`

Switches your working directory to the specified branch. Prefer `git switch` over `git checkout` for branch operations — it is more explicit and less error-prone.



Tip: Run `git remote show origin` before and after changing remotes to confirm the configuration is correct. Always follow with `git fetch --all` to update your local references.